Linhai Ma

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Summary

Machine learning engineer building large language model systems for technical domains, from model-level methods (prompt engineering, supervised fine-tuning, preference optimization) to production deployment (retrieval, inference serving, evaluation, reliability).

- Production LLM systems: led retrieval performance and reliability for an enterprise LLM question-answering service for chip design and verification, covering NVIDIA NIM inference on Kubernetes, retrieval redesign, HNSW tuning, and production debugging down to the C++ index layer.
- Structured generation and alignment: built pipelines that turn unstructured domain text into schema-valid records, and developed TAB-PO, a token-level preference optimization method.
- Retrieval, agents, and evaluation: proposed an evidence-to-taxonomy retrieval method over a 17K-concept inventory and a reference-derivation method (DeRef) for tool-call argument judges. Also built the auditing environment and evaluation harness for an agentic financial benchmark, run against five agent frameworks and four backbones.

Domain experience spans semiconductor design automation (three engagements at Cadence Design Systems between 2021 and 2024), clinical NLP (Yale University, with clinicians at Yale–New Haven Hospital and Cleveland Clinic), financial AI (The Fin AI Community), and adversarial robustness in medical imaging (PhD work).

Education

University of Miami, Ph.D. in Computer Science, Coral Gables, FL 2017 – 2023

- GPA: 3.85/4.0
- **Research Interest:** Robustness of deep neural networks, Medical Image/Signal Analysis

Institute of Software, Chinese Academy of Sciences, MS in Computer Science, Beijing, China 2014 – 2017

- GPA: Top 10%
- **Research Interest:** Concurrent Software Testing

Northeastern University, BS in Information Security, Shenyang, China 2010 – 2014

- GPA: Top 5%
- **Coursework:** C\C++, Java, Data Structure, Computer Architecture, Operating System, Computer Network, etc.

Programming Skills

- **Programming Language:** Python, C/C++, Java, C#, Ruby, R, SQL, Prolog, etc.
- **Framework:** Pytorch, Pandas, Scikit-learn, Numpy, Kubernetes, Docker, PyQT5, Java Swing, etc.

Experience

Postdoctoral Associate, Yale University – New Haven, CT Jan 2025 – Present

- Built the extraction pipeline that turns free-text patient–provider portal messages into schema-valid records, each required to carry its semantic labels and an evidence span anchored in the source text.
- Designed a field-level evaluation that scores labels and evidence separately, so wrong-evidence and wrong-label failures can be told apart, and released it as an annotated benchmark for hierarchical clinical extraction.
- Developed and implemented TAB-PO, a token-level preference optimization method that intervenes only at decoding positions below a confidence threshold.
- Ran the fine-tuning and preference-optimization on LLMs from 1.5B to 70B.

Research Collaborator, The Fin AI Community May 2026 – Present

- Proposed Factorized Hypothesis Search (FHS) for evidence-to-taxonomy retrieval, which combines factorized hypotheses, rank fusion, and an LLM verifier, and evaluated it on 17K-concept US-GAAP tagging and ICD-10 clinical coding.
- Found that tool-call argument judges accept correct values far less often when their input gives no explicit reference for them, across seven judges and three benchmarks. Proposed DeRef, which derives a reference from the full history without seeing the candidate and improves argument AUROC for every judge tested.
- Built the MCP-grounded auditing environment and evaluation harness for Herculean, an agentic financial benchmark, run against five agent frameworks and four backbones.

Machine Learning Engineer, Cadence Design Systems – San Jose, CA Jul 2024 – Dec 2024

- Led retrieval performance and reliability for the LLM question-answering service in Cadence JedAI, the company's enterprise AI platform for chip design and verification, serving internal and external EDA engineers over tens of thousands of documents.
- Cut retrieval latency from $\sim$10s to $\sim$100ms by fixing a filter-pushdown mismatch that made the vector store bypass its HNSW index and scan most of the corpus on every query.
- Tuned HNSW parameters and embedding models as controlled recall-latency experiments, evaluated on a validation set labeled by domain-expert application engineers.
- Resolved a production outage caused by a native-memory leak in the vector store's C++ HNSW layer, and deployed NVIDIA NIM inference microservices on Kubernetes with NVIDIA engineering.

Postdoctoral Associate, University of Florida – Gainesville, FL Jan 2024 – Jun 2024

- Built a multi-modal deep learning system for clinical decision support in prostate tumor diagnosis, integrating medical imaging (micro-ultrasound) with structured clinical variables from patient records.
- Designed a CNN-based classification model for cervical vertebrae maturation (CVM) stage assessment from medical images, achieving 76.7% accuracy.
- Developed generative models (diffusion-based) for cross-modality medical image translation, enabling data augmentation and domain adaptation across imaging protocols.

Research Assistant, University of Miami – Coral Gables, FL Sep 2019 – Dec 2023

- Developed robust deep learning methods for medical image analysis, including MR image segmentation, X-ray landmark detection, and blood cell object detection.
- Designed adaptive adversarial training algorithms with sample-specific margin adjustment to improve model robustness on medical imaging tasks while preserving classification and segmentation accuracy.
- Proposed a regularization-based framework to enhance CNN robustness for ECG signal classification under noisy conditions.
- Systematically evaluated robustness methods across diverse medical imaging modalities and architectures, demonstrating consistent generalization across clinical vision tasks.

Intern Software Engineer in Machine Learning, Cadence Design Systems, Inc. – San Jose, CA May 2022 – Aug 2022

- Designed and implemented a machine learning system to accelerate worst-case measurement evaluation of integrated circuits (ICs), reducing discovery time by over 90% compared to traditional simulation-based methods.
- Built Gaussian Process regression models in Python/PyTorch, trained on simulation data sampled via Latin Hypercube Sampling.
- Applied Bayesian optimization to iteratively select simulation inputs most likely to produce worst-case measurements, dynamically updating the training dataset.
- Performed forward feature selection to remove noisy input features and reduce simulation outliers.

Intern Software Engineer in Machine Learning, Cadence Design Systems, Inc. – San Jose, CA May 2021 – Aug 2021

- Designed a machine learning system to predict integrated circuit (IC) simulation outputs using Python and Scikit-learn.
- Built a data pipeline to preprocess historical simulation data into structured training sets.
- Implemented a Random Forest model to handle highly diverse simulation input ranges, achieving a mean absolute error of 0.99 picoseconds.
- Developed an interactive UI with PyQt5 and Matplotlib to visualize feature impact on simulation outcomes.

Research Assistant, University of Miami and Northwestern University – Coral Gables, FL — Jan 2018 – Aug 2019

- Identified evolution patterns of successful academic groups through large-scale data mining.
- Parsed author and publication records (1991–2018) from the Microsoft Graph database using SQL, and constructed million-scale annual coauthor networks with Python and NetworkX.
- Designed a Monte Carlo-based clustering algorithm in Python and R to detect author groups within networks.
- Defined group evolution patterns (e.g., splitting, merging) across adjacent years.
- Analyzed the correlation between group evolution and academic success (e.g., citation counts) using statistical methods in Python and R.

Research Assistant, Institute of Software, Chinese Academy of Sciences – Beijing, China — Sep 2015 – May 2017

- Developed a multi-threaded C++ test case generator for validating concurrent data structures.
- Proposed and implemented three adaptive generation strategies, increasing concurrency error detection by up to 6% while reducing execution time by up to 10%.

Intern Software Engineer (C/C++), Hillstone Networks – Beijing, China — Jul 2015 – Nov 2015

Intern Software Engineer (Java), Huaxin Education Technology Co., Ltd. – Shenyang, China — Jun 2013 – Aug 2013

Selected Review Experience

- NeurIPS (2022 – 2025)
- ICML (2022, 2025, 2026)
- ICLR (2024 – 2026)
- AISTATS (2026)
- AAAI (2026)
- AI
- Neurocomputing

Teaching Experience

Teaching Assistant, CB&B 750 01 (SP25): Core Topics in Biomedical Informatics – Yale University — Spring 2025

Teaching Assistant, CSC546: Introduction to Machine Learning – University of Miami — Spring 2023

Teaching Assistant, CSC315: Introduction to Python for Scientists – University of Miami — Spring 2023

Teaching Assistant, CSC546: Introduction to Machine Learning – University of Miami — Fall 2022

Teaching Assistant, CSC315: Introduction to Python for Scientists – University of Miami — Spring 2020

Teaching Assistant, CSC315: Introduction to Python for Scientists – University of Miami — Fall 2019

Teaching Assistant, CSC546: Introduction to Machine Learning – University of Miami — Spring 2019

Teaching Assistant, CSC421: Computer Operating Systems – University of Miami — Fall 2018

Teaching Assistant, CSC421: Computer Operating Systems – University of Miami — Fall 2017

Publications

Judges Follow the Reference: The Derivation Gap in Verifying Tool-Call Arguments — Under Review

L. Ma, Y. Wang, L. Qian, Y. Cao, V. Gutiérrez-Basulto

https://arxiv.org/pdf/2610.18497

Factorized Hypothesis Search for Evidence-to-Taxonomy Retrieval — Under Review

L. Ma, E. F. Wei, X. Peng, Y. Wang, L. Qian, V. Gutiérrez-Basulto

https://arxiv.org/pdf/2608.06614

TAB-PO: Preference Optimization with a Token-Level Adaptive Barrier for Token-Critical Structured Generation — Under Review

S. Fodeh, ***L. Ma***, G. Puthiaraju, S. Talakokkul, A. Khan, A. Hagaman, S. Lowe, A. Roundtree

https://arxiv.org/pdf/2603.00025

PVminerLLM: Structured Extraction of Patient Voice from Patient-Generated Text using Large Language Models — 2026

S. Fodeh, ***L. Ma***, G. Puthiaraju, S. Talakokkul, A. Khan, A. Hagaman, S. Lowe, A. Roundtree

NPJ Digital Public Health, Accepted

EPPCMinerBen: A Novel Benchmark for Evaluating Large Language Models on Electronic Patient-Provider Communication via the Patient Portal — 2026

S. Fodeh, Y. Wang, ***L. Ma***, S. Talakokkul, J. Alpert, S. Schellhorn

Artificial Intelligence in Medicine

PVminer: A Domain-Specific Tool to Detect the Patient Voice in Patient Generated Data — 2026

S. Fodeh, ***L Ma***, Y Wang, S Talakokkul, G. Puthiaraju, A. Khan, A. Hagaman, S. Lowe, A. Roundtree

Scientific Report, Accepted

EPPC-OASIS: Ontology-Aware Adaptation and Structured Inference Refinement for Electronic Patient-Provider Communication Mining in Secure Messages — Under Review

S. Fodeh, S. Ramachandran, E. Irankhah, M. Arif, A. Khan, G. Puthiaraju, **L. Ma**, S. Talakokkul, J. Alpert, S. Schellhorn

https://arxiv.org/pdf/2605.24172

Herculean: An Agentic Benchmark for Financial Intelligence — Under Review

X. Peng, Z. Xie, Y. Cao, H. Li, L. Qian, Y. Wang, V. J. Zhang, H. He, X. Ai, **L. Ma**, et al. (60 authors)

https://arxiv.org/pdf/2605.14355

STaR-DRO: Stateful Tsallis Reweighting for Group-Robust Structured Prediction — Under Review

S. Fodeh, G. Puthiaraju, E. Irankhah, ***L. Ma***, S. Talakokkul, A. Khan, S. Ramachandran, J. Alpert, S. Schellhorn

https://arxiv.org/pdf/2604.09737

Attention-based Shape-Deformation Networks for Artifact-Free Geometry Reconstruction of Lumbar Spine from MR Images 2025

L Qian, J Chen, ***L Ma***, T Urakov, W Gu, L Liang

IEEE Transactions on Medical Imaging

Intrathread Method Orders Based Adaptive Testing of Concurrent Objects 2025

Y Dai, L Xie, P Wu, S Cui, ***L Ma***

Science of Computer Programming, 103362

A novel attention-based network for geometry reconstruction with error estimation from medical images 2025

L Qian, J Chen, ***L Ma***, T Urakov, W Gu, L Liang

Medical Imaging 2025: Image Processing 13406, 169–176

Symtc: A symbiotic transformer-cnn net for instance segmentation of lumbar spine MRI 2024

J Chen, L Qian, ***L Ma***, T Urakov, W Gu, L Liang

Computers in Biology and Medicine, 179, 108795

Adaptive Testing of Concurrent Objects 2024

Y Dai, P Wu, S Cui, ***L Ma***

Theoretical Aspects of Software Engineering (TASE 2024), Guiyang, China

Evaluation of Deep Neural Network Models for Instance Segmentation of Lumbar Spine MRI 2024

J Chen, L Qian, ***L Ma***, T Urakov, W Gu, L Liang

bioRxiv, 2024.04.02.587810

A sequential geometry reconstruction based deep learning approach to improve accuracy and consistence of lumbar spine MRI image segmentation 2024

L Qian, J Chen, ***L Ma***, T Urakov, L Liang

Medical Imaging 2024: Image Processing 12926, 753–760

A general approach to improve adversarial robustness of DNNs for medical image segmentation and detection 2024

L Ma, J Chen, L Qian, L Liang

Medical Imaging 2024: Image Processing 12926, 266–274

Towards Improving the Adversarial Robustness of Deep Neural Networks 2023

L Ma

University of Miami Dissertation

Improving Adversarial Robustness of Deep Neural Networks via Adaptive Margin Evolution 2023

L Ma, L Liang

Neurocomputing, 551, 126524

Increasing-Margin Adversarial (IMA) Training to Improve Adversarial Robustness of Neural Networks 2023

L Ma, L Liang

Computer Methods and Programs in Biomedicine, 240, 107687

Towards lifting the trade-off between accuracy and adversarial robustness of deep neural networks for medical image classification and segmentation 2023

L Ma, L Liang

Medical Imaging 2023: Image Processing 12464, 429–436

Adaptive Adversarial Training to Improve Adversarial Robustness of DNNs for Medical Image Segmentation and Detection 2023

L Ma, L Liang

arXiv:2206.01736

A Regularization Method to Improve Adversarial Robustness of Neural Networks for ECG Signal Classification 2022

L Ma, L Liang

Computers in Biology and Medicine, 144, 105345

Enhance CNN Robustness Against Noises for Classification of 12-Lead ECG with Variable Length 2020

L Ma, L Liang

19th IEEE International Conference on Machine Learning and Applications (ICMLA 2020)

An Algorithm for Out-Of-Distribution Attack to Neural Network Encoder 2020

L Liang, ***L Ma***, L Qian, J Chen

arXiv:2009.08016

Improve robustness of DNN for ECG signal classification: a noise-to-signal ratio perspective 2020

L Ma, L Liang

ICLR 2020 Workshop on AI for Affordable Health

Diversity driven adaptive test generation for concurrent data structures 2018

L Ma, P Wu, T.Y. Chen

Information and Software Technology, 103, 162–173